

Unit: II), Sequential Model, Queuing Theory and Related Problems.

* Processing n jobs through 2 machines:-

Steps:- 1) List all the jobs and their processing time on machine M_1 and machine M_2 in a table.

2) Look at all the processing times on both machines (M_1 and M_2) and find the smallest processing time.

3) Case:1) If the smallest time is on machine M_1 , place that job at the beginning of the sequence (first available position).

4) Case:2) If the smallest time is on machine M_2 , place that job at the end of the sequence (last available position).

4) Remove the assigned job from the table.

Repeat step 2 and 3 for the remaining jobs until all jobs are arranged.

5) After getting the final sequence, calculate,

- Total elapsed time
- Idle time for machine M_1
- Idle time for machine M_2

• If tie occurs:- (Case:1) If the minimum time is same on M_1 and M_2 for a job, then select that job first

2) If the same smallest time occurs for two or more jobs on M_1 , then choose the job with smallest job number.

3) If the smallest time occurs for two or more jobs on M_2 , then, choose the job with the largest job no, place it at the end of the sequence.

Ex-1) There are 7 jobs, each of which has go through the machine A and machine B in order. AB processing time in hours as follows.

Job	1	2	3	4	5	6	7
machine A	3	12	15	6	10	11	9
machine B	8	10	10	6	12	1	3

Determine the sequence of these jobs that will minimize the total elapsed time. Also find the Idle time for machine A and B.

Ans:- Here, Optimal sequence of jobs,

1	4	5	2	3	7	6
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Now, the total Elapsed time is,

Job	Machine A		Machine B		Idle time for B.
	In time	out time	In time	Out time	
1	0	$0+3=3$	3	$3+8=11$	03
4		$3+6=9$	11	$11+6=17$	00
5	9	$9+10=19$	19	$19+12=31$	02
2	19	$19+12=31$	31	$31+10=41$	00
3	31	$31+15=46$	46	$46+10=56$	05
7	46	$46+9=55$	56	$56+3=59$	00
6	55	$55+11=66$	66	$66+1=67$	07

Total elapsed time to complete all the jobs is 67 hours.

Idle time for machine A is $= 67 - 66 = 1$ hour.

Idle time for machine B is $= 17$ hours.

Basic Terminology :-

- 1) Processing time :- This is the time required by a job on each machine.
- 2) Processing order :- This refers to a sequence in which machine are required for completing the job.
- 3) Idle time on a machine :- This is the time during which a machine does not have a job to process.
- 4) Total elapsed (gone) time :- This is the time interval between starting the first job and completing the last job including the idle time (if any), in a particular order/sequence by the given set of machines.
- 5) No passing Rule :- This rule of maintaining the order in which jobs are to be processed on given machines.

Processing n jobs through 3 machines :-

To find optimal Sequence of jobs, use Johnson's Algorithm,

• Johnson's Algorithm :-

- a) find minimum processing time on M_1 and M_3 and also maximum processing time on M_2 .
- b) Verify, $\min\{\text{Machine } M_1\} \geq \max\{\text{Machine } M_2\}$ or $\min\{\text{Machine } M_3\} \geq \max\{\text{Machine } M_2\}$
- c) If any one condition is satisfied, then take,
machine A = $M_1 + M_2$
machine B = $M_2 + M_3$.

* Steps to solve :-

- 1) Find optimum Sequence by using Johnson's Rule.
- 2) Solve by using previous examples.

Q. There are 7 jobs each of which must go through three machines M_1, M_2, M_3 in order M_1, M_2, M_3 . The corresponding processing times, on machines in hours are as follows;

Job	1	2	3	4	5	6	7
M_1	3	0	8	7	4	9	8
M_2	4	3	2	5	1	4	3
M_3	6	7	5	11	5	6	12

Determine the optimal sequence of jobs that will minimize the total elapsed time. Also, find the idle time for machine M_1, M_2, M_3 .

Soln:- Here, $\min\{M_1\} = 3$, $\min\{M_3\} = 6$, $\max\{M_2\} = 5$.

So, $\min\{M_3\} \geq \max\{M_2\}$.

So, by Johnson's Rule,

$$A = M_1 + M_2, \quad B = M_2 + M_3.$$

$\therefore$ Jobs 1 2 3 4 5 6 7

A- 7 11 9 9 10 12 16

B 16 10 7 16 6 16 15

Now, the optimal seqⁿ is,

1	4	7	6	3	5
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Now, to find total elapsed times,

Job	machine M_1		Machine M_2		Machine M_3		Idle o. M_3
	Start time	finish time	start	finish	Start	finish	
1	0	$0+3=3$	$3+4=7$	$3+4=7$	7	$7+6=13$	
4	3	$3+4=7$	7	$7+5=12$	13	$13+11=24$	
7	7	$7+7=14$	14	$14+3=17$	24	$24+12=36$	
6	14	$14+8=22$	22	$22+4=26$	36	$36+6=42$	
2	22	$22+8=30$	30	$30+3=33$	42	$42+7=49$	
3	30	$30+7=37$	37	$37+2=39$	49	$49+5=54$	
5	37	$37+9=46$	46	$46+1=47$	54	$54+5=59$	

∴ Total elapsed time = 59 hours.

The Idle time on $M_1 = 59 - 46 = 13$ hours.

Idle time on $M_2 = 3 + 0 + 2 + 5 + 4 + 4 + 7 + (59 - 47)$
 $= 37$ hours.

Idle time on $M_3 = 7 + 0 + 0 + 0 + 0 + 0 + 0$
 $= 7$ hours.

- 2) A Publisher propose to publish five different books. Each book process through three steps composing, printing and binding. The time taken by each of these is given below.

Book	1	2	3	4	5
Composing	40	90	80	60	50
Printing	50	60	20	30	40
Binding	80	100	60	70	110

Determine the optimum sequence of books, also find idle time for each.

Soln:- Here, min of composing = 40
 (C)

max of Printing = 60
 (P)

min of binding = 60

Binding (B)	60	7	4	1
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Here, min of B $\geq$ max of P. Lot of book will

We use Johnson's algorithm here, M machine

$M_1 = C + P$; $M_2 = P + B$ mit that

∴ We get,

Book	1	2	3	4	5
$M_1 = C + P$	90	150	100	90	90
$M_2 = P + B$	130	160	80	100	150

Now, to find the optimum Sequence,

4	1	5	2	3
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Also, to find total elapsed time and idle time for each machine,

Book	Composing		Printing		Binding	
	In	out	In	out	In	out
4	0	$0+60=60$	60	$60+30=90$	90	$90+70=160$
1	60	$60+40=100$	100	$100+50=150$	160	$160+80=240$
5	100	$100+50=150$	150	$150+40=190$	240	$240+110=350$
2	150	$150+90=240$	240	$240+60=300$	350	$350+100=450$
3	240	$240+80=320$	320	$320+20=340$	450	$450+60=510$

Here, Total elapsed time = $510 - 0 = 510$ hours.

Idle time for composing = $510 - 360$

= 190 hours.

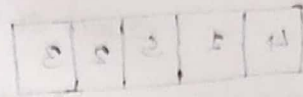
Idle time for printing = $60 + 10 + 0 + 50 + 20 + 130$

= 270 hours.

Idle time for binding = $90 + 0$

= 90 hours.

Queue Theory :-



• Introduction :- Queue Theory is the study of waiting lines. It helps us analyse situations where customers arrive, wait for service, get served and then leave. The main aim is to provide good service with minimum waiting time.

• Objective of Queue Analysis :-
Queue analysis is used to -
1) Reduce waiting time of customers,
2) Improve service efficiency.
3) Use service facilities properly.
(machine, workers, counters, etc).

* Characteristics of Queue Theory :-

1) Input process or Arrival Pattern :-

The customers (machines, people workers, etc) arrive and join the system through an input process customer arrive at the system for service.

2) Queue or waiting line :-

a) waiting time of a customer in queue :- It is a time spend by a customer in the queue before being served.

It is denoted by W_q .

b) Waiting time of a customer in the system :- It is the total time spend by the customer in the system.

i.e. it is the time which is spend in queue and service time. It is denoted by W_s .

c) Queue length :- It is the number of customers waiting for service in the queue.

It is denoted by L_q .

d) System length :- It is the number of customers in the system. i.e., those in queue plus those being served.

It is denoted by L_s .

* Important formulae:-

- 1) Probability that the service facility is busy.
i.e. the probability that there is atleast one customer in the system is given by,

$$\rho = \frac{\lambda}{\mu} \quad (\rho = \text{ratio})$$

Where, λ = Mean arrival rate

μ = Mean service rate.

- 2) Probability that the service facility is idle (free). i.e. there is no customer in the system.

$$P_0 = 1 - \rho = 1 - \frac{\lambda}{\mu}$$

- 3) Probability that there are n -customers in the system.
i.e. the customer those are being served and those are waiting in the queue

$$P_n = P_0 \left(\frac{\lambda}{\mu} \right)^n = \left(1 - \frac{\lambda}{\mu} \right) \left(\frac{\lambda}{\mu} \right)^n$$

- 4) Average waiting time of customers in the system. i.e. waiting time of a customer in the queue and its service time.

$$W_s = \frac{1}{\mu - \lambda}$$

- 5) Average waiting time of a customer in the queue

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{\lambda}{\mu} \cdot W_s$$

- 6) Average or Expected number of customers in the system; i.e. system length.

$$L_s = \frac{\lambda}{\mu - \lambda} = \lambda \cdot W_s$$

- 7) Average or Expected no. of customers in the queue i.e. queue length.

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \lambda W_q = \frac{\lambda}{\mu} \cdot L_s$$

- 8) Probability that waiting time of a customer is more than t hours.

$$\text{In the system} = e^{-(\mu - \lambda)t}$$

$$\text{In the queue} = \frac{\lambda}{\mu} \times e^{-(\mu - \lambda)t}$$

- 9) Probability that queue size (length) is more than or equal to k i.e. probability that number of customers in system exceeds k ,

$$P(n > k) = \left(\frac{\lambda}{\mu}\right)^{(k+1)}$$

- 10) Average number of customers in the non-empty queue i.e. the queue length if it forms time to time

$$L_w = \frac{\lambda}{\mu - \lambda}$$

Que. 1) In a railway marshalling yards, goods trains arrive at a rate of 30 trains per day. Assuming that the inter arrival time follows poisson distribution and the service time distribution follows an exponential distribution with an average of 36 min.

Calculate the following-

- 1) Mean queue size
- 2) Average time the train spend in the yard.
- 3) Probability that the queue size exceeds 10.
- 4) Probability that the arriving train does not have to wait.

Soln:- Given that,

Mean arrival rate of trains (λ) = 30 per day.

Mean service rate μ = 36 min/train

$$= \frac{24 \times 60}{36}$$

$$= 40 \text{ per day.}$$

Here,

① Mean queue size:

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{(30)^2}{40(40 - 30)}$$

$$= 2.25 \text{ train.}$$

② Average time the train spend in the yard,

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{10} \text{ per day}$$



③ probability that queue size exceeds 10

$$\begin{aligned} \text{i.e. } P(n > 11) &= \left(\frac{\lambda}{\mu}\right)^{(k+1)} \\ &= \left(\frac{30}{40}\right)^{(11+1)} = \left(\frac{30}{40}\right)^{12} \\ &= 0.031 \end{aligned}$$

④ probability that the arriving train does not have to wait
i.e. yard is idle.

$$\begin{aligned} P_0 &= 1 - \frac{\lambda}{\mu} = 1 - \frac{30}{40} \\ &= 0.25 \end{aligned}$$

Q.2) A departmental store has one cashier is there to serve the customers. And the customers pick up their needs by themselves. The arrival rate is 20 customers per hour and cashier can serve 24 customer per hour. Assume poisson arrival rate and exponential distribution for service rate find

- ① Average number customers in the service.
- ② Average time a customer spend in the system.
- ③ Average time customer spend in the queue.

Soln:- Given that,

Mean arrival rate (λ) = 20 customer per hour.

Mean Service rate (μ) = 24 customer per hour.

① Average no. of customers in the system:

$$\begin{aligned} L_s &= \frac{\lambda}{\mu - \lambda} = \frac{20}{24 - 20} \\ &= 5 \text{ customer} \end{aligned}$$

② Average time customer spend in the system:

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{24 - 20} = \frac{1}{4} \text{ hour.}$$



$$= \frac{60}{4} \text{ min}$$

$$= 15 \text{ min.}$$

③ Average time customer spend in the queue

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)}$$

$$= \frac{20}{24(24 - 20)} = \frac{20}{24 \times 4} = \frac{5}{24} \text{ hours.}$$

$$= \frac{5 \times 60}{24} \text{ min.}$$

$$= 12.5 \text{ min.}$$